# Combining SAM With Limited Data for Change Detection in Remote Sensing

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***Abstract*— Change detection is a critical task in the remote sensing image (RSI) analysis, widely used in fields such as land cover change and urban planning. With the introduction of foundational models like SAM in computer vision (CV) tasks, their advantages in zero-shot and interactive segmentation have enabled rapid application across diverse visual scenarios. Current research in change detection focuses on designing learnable plug-in modules and fine-tuning foundational models using large annotated data. However, constructing comprehensive datasets and designing effective additional modules pose significant challenges, leading to high costs. To address these issues, we propose a model named Meta-CD for remote sensing change detection (RSCD) with limited data. By introducing a simple fine-tuning module, this model is trained on limited datasets and quickly adapts to change detection tasks. Specifically, we integrate an additional CNN as an adapter with the foundational model FastSAM. Initially, we freeze the parameters of FastSAM and train only the parameters of the introduced adapter and decoder to generate change confidence maps. Subsequently, to enhance the quality of change detection, we introduce a novel pixel-level binarization module that learns the threshold for each pixel in the original image. This module combines the thresholds with the confidence maps to output binary change detection maps, filtering out invalid change pixels. Experimental results demonstrate that our method outperforms other competing approaches on limited datasets and has great zero-shot learning ability. Our code is available at Meta-CD.**

***Index Terms*— Change detection, foundation model, limited data, remote sensing.**

## I. Introduction

REMOTE sensing change detection (RSCD) involves comparing remote sensing images (RSIs) taken at different times to detect changes in surface features or objects over time [1]. With advancements in image processing technologies and increasingly sophisticated satellite observation systems, the application of remote sensing data in Earth science research has become more prevalent [2]. RSCD is a fundamental method in the remote sensing data analysis, enabling real-time monitoring of surface and object changes, thus providing a scientific basis for resource management and environmental protection [3]. Currently, RSCD technologies are widely applied across various fields, including forest vegetation detection [4], land cover change [5], environmental monitoring [6], disaster monitoring [7], and urban planning [8].

Over the past decade, convolutional neural networks (CNNs) have gained increasing popularity in various computer vision (CV) tasks [9]. Certain CNN techniques and modules for semantic segmentation have been applied to RSCD tasks to extract deeper features. Initially, U-Net established the benchmark model [10], and subsequently, Siamese networks became the standard approach for change detection [11]. With the rise of Transformers in visual tasks, Transformer-based methods have accelerated the progress of RSCD [12]. Compared with traditional CNNs, Transformers can capture long-range dependencies across entire images, providing models with a global receptive field and opening new avenues for tasks requiring advanced semantic understanding [13]. However, these methods generally require large amounts of training data, which is both expensive and time-consuming.

In recent years, foundation models have emerged as a new paradigm in deep learning, demonstrating exceptional performance in fields, such as natural language processing (NLP) [14]. In the CV domain, visual foundation models (VFMs), such as "segment anything model" (SAM) [15] and its variant FastSAM [16], have been applied to numerous tasks due to their robust feature extraction and generalization capabilities. Despite their significant success, these methods face substantial challenges in adapting to the complex and dynamic scenarios of RSCD, falling short of practical application needs. As model sizes increase, the limited availability of labeled data significantly undermines their potential. Although many researchers have developed various fine-tuning methods to optimize SAM for diverse downstream tasks in different domains—such as using adapter-based approaches or employing LoRA [17] and its variants—these methods typically require considerable effort and resources for model

Received 17 December 2024; revised 6 February 2025; accepted 20 February 2025. Date of publication 25 February 2025; date of current version 11 March 2025. This work was supported in part by the Natural Science Foundation of China under Grant 62306241 and in part by the National Key Research and Development Program of China under Grant 2022ZD0160400. *(Corresponding author: Zhiyuan Zhao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3545040

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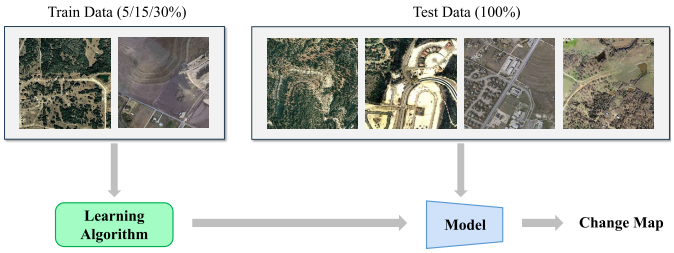



Fig. 1. Schematic of remote sensing with limited data. In some cases, the number of training samples is significantly smaller than the number of test samples.

training. Applying VFMs to RSCD with limited data remains an unresolved challenge. (Fig. 1 shows a schematic of RSCD with limited data.)

To address these challenges, we propose a limited-data RSCD model named Meta-CD, capable of rapidly performing RSCD tasks with constrained datasets. Unlike previous studies, our work introduces a novel adapter by integrating an additional CNN as a supplementary encoder into the SAM framework. This approach allows for flexible integration of supplementary networks while avoiding backpropagation through the entire VFM, thereby accelerating training and reducing resource costs. The additional CNN can be replaced by any form of encoder based on specific task requirements. In this study, we use a pretrained ResNet [9] as the supplementary network. During training, only the parameters of the additional CNN and decoder parts are updated, while the parameters of the original FastSAM encoder remain frozen.

In addition, we observe that the network exhibits varying confidence distributions for building changes at different scales. Existing methods typically use a fixed threshold to convert change confidence maps into binary images [18]. However, the goal of RSCD is to detect specific types of changes, rather than identifying all differences between observations. Using a fixed threshold can lead to missed boundaries of change areas and false positives in irrelevant regions, often resulting in poor performance. To accurately capture objects from regions with different confidence levels, we introduce a pointwise thresholding approach generated by a pixelwise binarization module (PBM). This module comprehends the distribution of semantic regions in RSI and computes distinct thresholds for each pixel during binarization. Consequently, PBM produces fine-grained binary images, achieving more precise change detection results than fixed thresholding. Meta-CD aims to leverage SAM's powerful and versatile semantic extraction capabilities to enhance RSCD accuracy and reduce dependence on large training samples. Our contributions can be summarized as follows.

1) We propose an adapter based on an additional CNN for fine-tuning to accommodate RSCD tasks. This adapter offers greater flexibility in designing supplementary networks while avoiding backpropagation through the VFM, minimizing resource costs.
2) We introduce a novel PBM that converts change confidence maps into binary images by understanding pixel-level semantic distributions in remote sensing imagery to learn large-scale changes.
3) Extensive experiments demonstrate that our model achieves competitive results with limited training samples and has great zero-shot learning ability.

## II. Related Works

In this section, we first review deep learning-based methods for RSCD in Section II-A. In Section II-B, we introduce some developments in VFMs, particularly focusing on SAM and its applications. Section II-C discusses some methods for learning with limited data.

### A. Deep Learning-Based Methods

In the field of RSCD, deep learning-based approaches have demonstrated significant potential and advantages [19]. Initially, CNN-based RSCD tasks were treated as segmentation problems [20], where models similar to U-Net [10] effectively integrated high-level and low-level features. Subsequently, the Siamese CNN framework [21] was introduced to extract multitemporal features by processing paired images in parallel through dual-branch networks [22]. Due to the self-attention mechanism of Transformer networks, which excels at handling long-range dependencies, researchers have leveraged its global receptive field to enhance feature refinement [12]. For instance, self-attention is employed in STANet [23], and dual attention is utilized in DASNet [24] to refine features and achieve superior feature representation. BiT [12] developed an efficient change detection model that combines CNNs and Transformers while maintaining a minimal parameter count. ChangeFormer [13] is a purely Transformer-based Siamese network that effectively extracts multiscale long-range details through a multistage Transformer encoder paired with a lightweight MLP decoder. Despite the notable success of these methods, their ability to adapt to complex and dynamic scenes still falls short of practical application requirements. As model sizes increase, the limited availability of annotated data significantly undermines the potential of these models.

### B. Vision Foundation Models

VFMs refer to models trained on large-scale datasets [25]. In the CV field, the concept of foundation models has gradually become key to addressing the challenges of data requirements and generalization in deep learning [26]. These models are pretrained on extensive datasets to acquire broad recognition capabilities, enabling them to adapt to various downstream tasks, particularly data-intensive ones like semantic segmentation [27] and CD [28]. CLIP [29], trained on a vast number of image–text pairs, has demonstrated performance in zero-shot image classification [30] tasks comparable to fully supervised CNNs, highlighting the potential of foundation models in handling user-prompted visual tasks. The SAM [15], trained on millions of annotated images, can segment unseen images and objects based on user prompts during inference. SegGPT [31] aims to learn visual concepts without explicit annotations by mimicking

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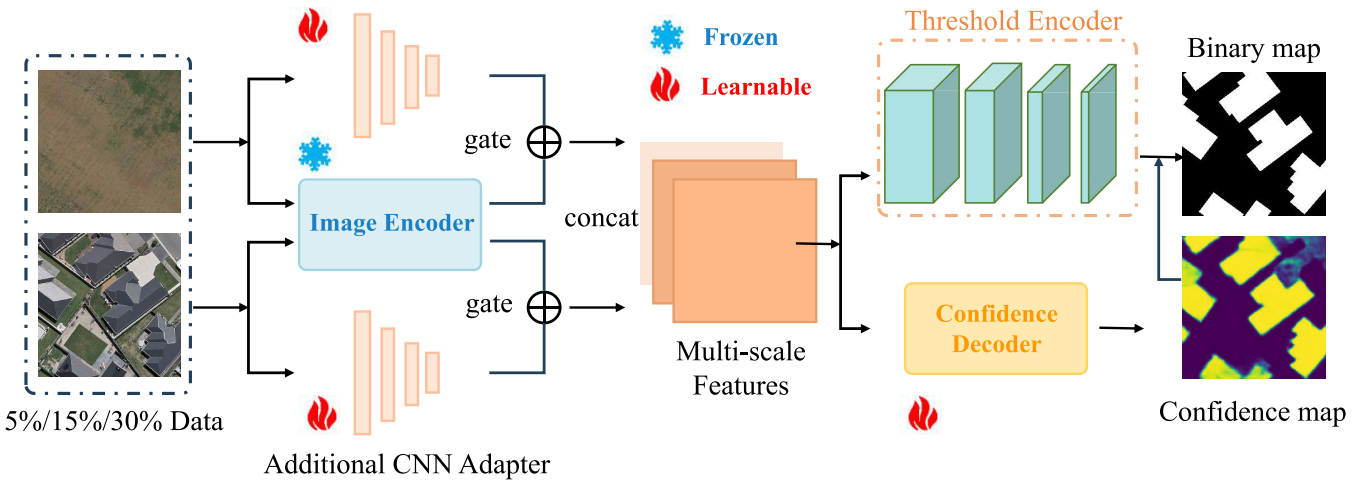



Fig. 2. Overview of our proposed framework for Meta-CD. Our method consists of two stages: change confidence map prediction and change detection map segmentation. In the first stage, we use the FastSAM encoder along with an additional CNN adapter to extract features from bitemporal images. These features are then used by our designed decoder to predict the change confidence map. In the second stage, pixel-level thresholds are learned to binarize the confidence map, resulting in the change detection map. In this process, the image encoder is the FastSAM encoder with its parameters frozen, while the remaining parameters are learnable.

human learning processes. FastSAM [16] seeks to achieve real-time segmentation, significantly enhancing inference speed while maintaining generalization performance comparable to SAM. However, VFMs face domain adaptation challenges when applied to RSI due to inductive biases learned from natural images [32]. To address this, this article introduces a fine-tuning method for VFMs to better learn and recognize latent semantic information in RSI.

### C. Learning With Limited Data

Limited data learning (LDL) is a technique that enables models to learn from a small number of labeled samples and generalize to new categories [33]. In LDL research, metalearning is a core approach that simulates limited data environments for rapid adaptation to new tasks. Model-agnostic metalearning (MAML) [34] is a representative work in metalearning, achieving quick adaptation to new tasks by learning model parameter initialization. In addition to MAML, metric-based methods also play a significant role in LDL, such as matching networks [35] and prototypical networks [36]. Integrating LDL with foundational vision models has become a crucial trend in deep learning [37], [38]. In this context, pretrained foundational vision models provide a robust starting point. Having learned rich feature representations from large datasets, they enable effective learning transfer from a few samples. This approach is particularly suitable for scenarios with scarce labeled data, such as medical image analysis [39] or specific industrial vision tasks [40]. In our study, our method is closely related to existing LDL work, but we opt for a simpler fine-tuning approach, effectively leveraging limited sample information to achieve excellent performance.

## III. Method

### A. Overview

Fig. 2 outlines the workflow of our method, which consists of two stages: change confidence map prediction and change detection map segmentation. The first stage involves feature extraction from the dual-temporal images using the FastSAM encoder and an additional CNN, followed by the prediction of the change confidence map using our designed decoder. The second stage focuses on learning pixelwise thresholds and binarizing the confidence map for change map segmentation.

### B. Additional CNN Tuning With FastSAM

FastSAM (see Fig. 3) is an innovative model designed to accelerate the "segment anything" task, addressing the high computational demands and real-time application limitations of SAM. It utilizes the CNN-based YOLOv8-seg model and divides the task into two stages [16]: all-instance segmentation and prompt-guided selection. The first stage generates segmentation masks for all instances, while the second stage selects specific objects based on prompts. FastSAM achieves performance comparable to SAM but operates 50 times faster, making it suitable for industrial applications requiring real-time processing. Despite the potential for improving mask quality and performance on small objects, FastSAM has demonstrated its efficiency and broad potential for practical applications.

To apply FastSAM's powerful feature extraction and generalization capabilities to the field of RSCD, we use FastSAM's encoder as the backbone network. The image encoder of FastSAM is based on the YOLOv8-x model, which performs feature extraction through a CNN and incorporates a feature pyramid network (FPN) to integrate multiscale features. Here, we describe our approach of introducing an additional convolutional network. Specifically, our design strategy involves appending a side network while keeping the parameters of the FastSAM model fixed, thus avoiding the influence of parameter updates from backpropagation. Our method only updates the parameters of the decoder and the additional CNN network, allowing us to fine-tune the FastSAM model on a limited building change detection dataset.

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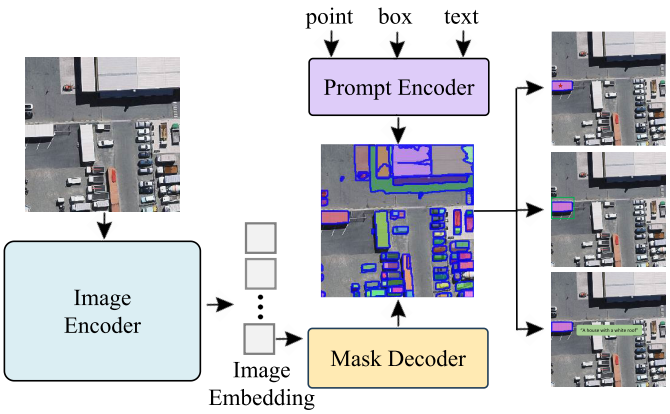



Fig. 3. Architecture of FastSAM. The model divides the "segment anything" task into two stages: first, a CNN detector generates segmentation masks for all instances in the image; then, specific objects of interest are selected based on prompts, such as points, boxes, or text.

Formally, given an input image $x$ ($x \in \mathbb{R}^{H\times W\times C}$), where $H$, $W$, and $C$ represent the height, width, and number of channels, respectively. This input image is simultaneously fed into the FastSAM encoder and the CNN encoder

$$F_{\text{sam}} = \text{FastSAM}_{\text{enc}}(x) \in \mathbb{R}^{H_i\times W_i\times C_i} \tag{1}$$

$$F_{\text{cnn}} = \text{CNN}_{\text{enc}}(x) \in \mathbb{R}^{H_i\times W_i\times C_i} \quad i \in [0, 1, 2, 3]. \tag{2}$$

The FastSAM encoder outputs multiscale features with four different channel numbers and sizes. The CNN encoder uses the basic architecture of ResNet. To ensure that the features extracted by the CNN have the same size as those extracted by the FastSAM encoder, we modify the number of output channels and output size of the four layers in ResNet, and output multiscale feature maps of the same size as FastSAM.

To quantify the contributions of the feature maps extracted by the two networks, we propose to introduce a learnable gating mechanism when fusing the feature maps of these two networks. This mechanism realized through a weight parameter $\alpha$, allows the model to adaptively adjust the influence of each feature map in the final fusion result

$$F = \alpha \cdot F_{\text{sam}} + (1-\alpha)\cdot F_{\text{cnn}}. \tag{3}$$

Specifically, this method weights the feature maps of the two networks using the introduced weight parameter $\alpha$ to generate a comprehensive feature representation through weighted summation. This weight parameter $\alpha$ is learnable, allowing the network to automatically optimize the fusion strategy of the feature maps during training. In this way, the model can flexibly capture information from different network feature maps and strengthen the feature representations that are critical for the target task.

In the decoder, we fuse the extracted multiscale features. We employ a decoder structure similar to U-Net to fuse adapted multiscale features, improving detection performance by combining results from different layers. The features at a smaller scale are enlarged through convolutional layers and then added to the extracted high-scale features for fusion. The specific scale transformation of image features is illustrated in Fig. 4. By employing this approach, the model can more effectively leverage the strengths of both the FastSAM and CNN encoders, leading to improved performance in applications such as RSCD. The use of the learnable gating mechanism ensures that the fusion process is dynamically optimized, enhancing the model's adaptability to varying data characteristics and tasks.

**Algorithm 1** Fusion of Additional CNN Features With FastSAM Features

**Input**: image $x \in \mathbb{R}^{H\times W\times C}$, weight parameter $\alpha$
**Output**: Fused feature map $F$
**Step 1: Feature Extraction**
$F_{\text{sam}} \leftarrow \text{FastSAM}_{\text{enc}}(x)$, $F_{\text{cnn}} \leftarrow \text{CNN}_{\text{enc}}(x)$
**Step 2: Modify ResNet Output**
**for** *each layer* $i$ *in* $\{0, 1, 2, 3\}$ **do**
  Modify output channels and size of ResNet layer $i$ to match FastSAM;
**end**
**Step 3: Feature Fusion**
Initialize learnable weight parameter $\alpha$
$F \leftarrow \alpha \cdot F_{\text{sam}} + (1-\alpha)\cdot F_{\text{cnn}}$
**Step 4: Feature Map Fusion in Decoder**
**for** *each scale* $i$ *in* $\{0, 1, 2, 3\}$ **do**
  Upsample smaller scale features;
  Add upsampled features to higher scale features;
**end**
**return** *Fused feature map* $F$

### C. Threshold Encoder

To focus on specific types of changes in RSCD, we filter out invalid change pixels and select valid ones by introducing a PBM that includes a threshold encoder. The threshold encoder processes the feature map $F \in \mathbb{R}^{H\times W\times C}$ to produce a threshold map $T$. Using a CNN, the encoder generates a threshold value for each pixel. The binarization layer then compares each pixel in the confidence map $I$ with the threshold map $T$ to produce the binary output $O$

$$O_{(i,j)} = \begin{cases} 1, & I_{(i,j)} \geq T_{(i,j)} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Since this process is nondifferentiable, we approximate the relationship between the binary output $O$ and the threshold $T$ using a linear function. Assume that $I \in \mathbb{R}^{H\times W}$ satisfies two conditions: 1) $I(i, j)$ follows a uniform distribution $U(0, 1)$ and 2) $H \times W$ approaches infinity. From a probabilistic perspective, we can compute the expected value $\hat{y}$ of the binary output $O$ to facilitate backpropagation

$$\hat{y} = \frac{1}{W\times H}\sum_{i\in H}\sum_{j\in W} O(i, j) = \begin{cases} 1, & T < 0 \\ 1-T, & 0 \leq T < 1 \\ 0, & T \geq 1. \end{cases} \tag{5}$$

During backpropagation, we update the threshold $T$ using the learning rate $\beta$ to ensure effective learning despite the

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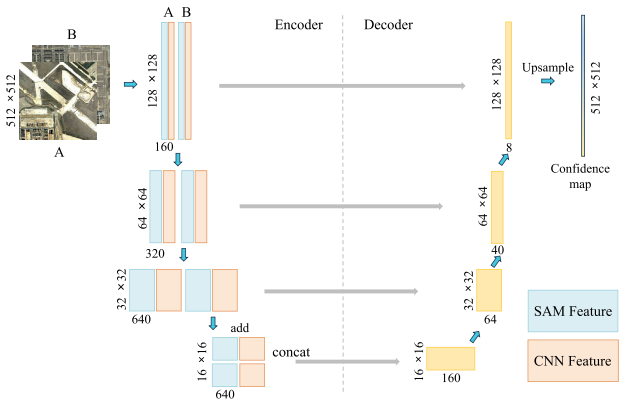



Fig. 4. Feature sizes and channel numbers obtained after passing the image through the encoder–decoder architecture.

nondifferentiable binarization

$$T = T + \beta \frac{\partial L}{\partial \hat{y}} \tag{6}$$

where $L(\hat{y}, y)$ is the loss function quantifying the difference between the predicted output $O$ and the ground truth (GT). This loss can be an $L1$ loss or any suitable metric.

### D. Loss Function

1) Confidence map learning is a regression problem, and according to our experiments, the BCE loss function has been effective in training it

$$L_{\text{conf}} = -(y \log(p) + (1 - y) \log(1 - p)). \tag{7}$$

2) Threshold map learning is a regression problem, and we choose the $L1$ loss function to train the threshold learner. We believe that threshold learning needs to converge faster than confidence prediction, so we did not use other loss functions

$$L_{\text{binar}} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|. \tag{8}$$

3) To improve the effectiveness of change detection, our model incorporates a semantic learning branch. Research [28] shows that unchanged regions should maintain semantic consistency, meaning that unchanged pixels should be as similar as possible in bitemporal features. We perform convolutional operations on the features extracted by the encoder to obtain latent semantic features $F_A$ and $F_B$. By aligning these feature representations, we ensure that the latent semantics of the change regions in the GT are significantly different, while the latent semantics of the unchanged regions remain similar. We use the cosine similarity function to calculate the semantic similarity between pre- and post-change images and define the semantic similarity loss $L_{\text{sim}}$ as follows:

$$L_{\text{sim}} = [1 - \text{Cosine}_{\text{sim}}(F_A, F_B)] \cdot (E - \text{GT}). \tag{9}$$

Here, GT represents the ground-truth change labels, and $E$ is an all-ones matrix of the same size as GT, used to exclude the loss computation for changed regions. $\text{Cosine}_{\text{sim}}$ is the cosine similarity function.

4) The total loss we obtain is the sum of the three losses

$$L_{\text{total}} = L_{\text{conf}} + L_{\text{binar}} + L_{\text{sim}}. \tag{10}$$

## IV. Experiments

### A. Datasets

*1) LEVIR-CD [23]:* It comprises very high-resolution (VHR) images from Google Earth, collected from six major Texas cities between 2012 and 2016, focusing on urban building expansion. The dataset includes 637 image pairs (1024 × 1024 pixels each), divided into training (445 pairs), validation (64 pairs), and testing sets (128 pairs). With a spatial resolution of 0.5 m/pixel, LEVIR-CD provides detailed annotations of 31 333 changed building instances.

*2) WHU-CD [41]:* Collected from Christchurch, New Zealand, between 2012 and 2016, it covers an area of 20.5 square kilometers with a high spatial resolution of 0.2 m/pixel. The original images are large, measuring 32 507 × 15 354 pixels, and have been divided into smaller 256 × 256 pixel blocks for easier handling RSCD task. The dataset is split into training (6096 pairs), validation (762 pairs), and testing sets (762 pairs).

*3) S2Looking-CD [42]:* It contains 5000 image pairs, each measuring 1024 × 1024 pixels. These pairs are divided into a 7:1:2 ratio for training, evaluation, and testing, respectively. The dataset includes over 65 920 instances of change annotation. The images, collected by optical satellites around the world, span a period of one to three years and have a resolution ranging from 0.5 to 0.8 m/pixel. For training and validation, these images are cropped to a size of 512 × 512 pixels.

### B. Evaluation Metrics

There are some general evaluation metrics for detection tasks, illustrating as follows:

$$\begin{aligned} P &= \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R = \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \quad F1 = \frac{2 \times P \times R}{P + R} \\ \text{OA} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \end{aligned} \tag{11}$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives, respectively. $P$ and $R$ represent the precision and recall of the detection. IoU is the intersection over union between the predicted bounding box and the GT. To consider both precision and recall simultaneously, the $F1$ score is calculated. OA stands for overall accuracy, which is used to measure the overall degree of consistency between the model's prediction results and the actual situation.

### C. Implementation Details

Meta-CD utilizes the pretrained FastSAM-x as the foundation. The original FastSAM parameters are frozen, while the other parameters in the network are trained. We use SGD optimizer to optimize the trainable parameters in the

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TABLE I
QUANTITATIVE RESULTS OF THE ABLATION STUDY ON 5% LEVIR DATASET

| Variants | Methods | Adapter | PBM | Computation Cost | | | Metrics | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Parameters(M) | Flops(G) | Inference(FPS) | OA | mIoU | F1 |
| 1 | Siamese ResNet(Baseline) | × | × | 2.74 | 8.77 | 10.56 | 96.81 | 59.42 | 69.51 |
| 2 | FastSAM | × | × | 4.71 | 8.95 | 6.62 | 97.39 | 76.97 | 72.31 |
| 3 | FastSAM+ResNet18 | ✓ | × | 13.66 | 10.37 | 6.55 | 97.70 | 78.28 | 74.17 |
| 4 | FastSAM+ResNet18 | ✓ | ✓ | 13.67 | 10.42 | 5.64 | **98.03** | **82.08** | **79.67** |
| 5 | FastSAM+ResNet34 | ✓ | ✓ | 23.77 | 14.05 | 3.13 | 97.96 | 81.42 | 78.76 |

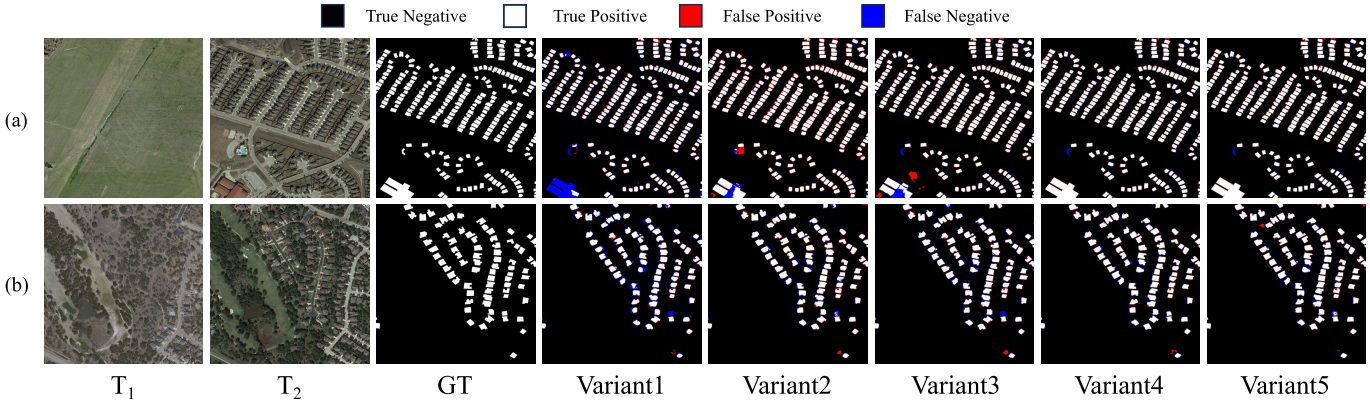



Fig. 5. Visualization of different variants in the ablation study. (a) and (b) Obtained on two test RSIs in the LEVIR-CD dataset.

proposed Meta-CD. The initial learning rates are set to 0.1 for the confidence predictor and 0.01 for the threshold encoder, and updated by multiplying with a scale factor related to the training iteration. Learning rate reduction helps Meta-CD achieve better change detection performance. We train on the LEVIR-CD and S2Looking datasets with a batch size of 4 and on the WHU-CD dataset with a batch size of 64, ensuring training on a single GPU (NVIDIA RTX3090). We trained Meta-CD for a total of 100 epochs and selected the top-performing model from the validation set for subsequent testing and evaluation.

### D. Ablation Study

The purpose of the ablation study is to explore the factors contributing to the significant performance of Meta-CD. To quantitatively assess the improvements brought by the proposed techniques, we compared the achieved accuracy with a baseline method. The baseline is constructed based on common settings in RSCD, utilizing two Siamese CNNs to extract temporal features, followed by adjustments. We conducted experiments on 5% of the LEVIR-CD dataset, with results shown in Table I. It can be observed that replacing the baseline method (Variant 1) with the simpler FastSAM (Variant 2) leads to a significant improvement in accuracy. OA, mean IoU (mIoU), and $F1$ score increased by 0.58%, 17.55%, and 2.80%, respectively. We also present visual examples of the change detection results in Fig. 5. Different colors are used to illustrate the key and detailed results of various variants, with blue and red indicating missed detections and false detections, respectively. Specifically, Variant 1, which uses only the twin ResNet, clearly exhibits missed and false detections and shows insensitivity to changes in the region. After incorporating FastSAM, Variant 2 shows a significant reduction in red and blue pixels. With the introduction of all modules, the techniques proposed by Meta-CD progressively improve segmentation results and enhance the ability to detect subtle changes.

*1) Impact of Additional CNN Adapter:* To evaluate the effect of additional CNN adapters, we compared the performance of FastSAM with and without these adapters. The results indicate that FastSAM with the additional CNN adapter (Variant 3) outperforms in all metrics, with improvements of 0.31%, 1.31%, and 1.86% in OA, mIoU, and $F1$ score, respectively, compared with Variant 2. The additional CNN adapter is designed to capture supplementary features that the FastSAM encoder might not fully extract. By integrating these extra convolutional layers, the network benefits from a more robust and comprehensive feature extraction process. This is particularly important in RSCD, where slight pixel value changes can indicate significant ground changes. Moreover, the additional CNN adapter refines feature maps, leading to more accurate segmentation than using the FastSAM encoder alone. The CNN adapter can also be replaced with other learnable feature extractors, such as ResNet34. Our experiments confirm that a network integrated with ResNet34 (Variant 5) performs slightly less than one with ResNet18 (Variant 4). This suggests that the choice of CNN backbone can be tailored to the specific requirements of the task, providing flexibility in model design. To balance the model efficiency and performance, we adopted Variant 4 in this study.

*2) Effectiveness of PBM:* PBM is another key component of the Meta-CD approach. Traditional change detection methods often use fixed thresholds to binarize change confidence maps. However, a fixed threshold may not be suitable for all regions of an image, leading to suboptimal segmentation results. To address this issue, we introduce a learnable threshold in the

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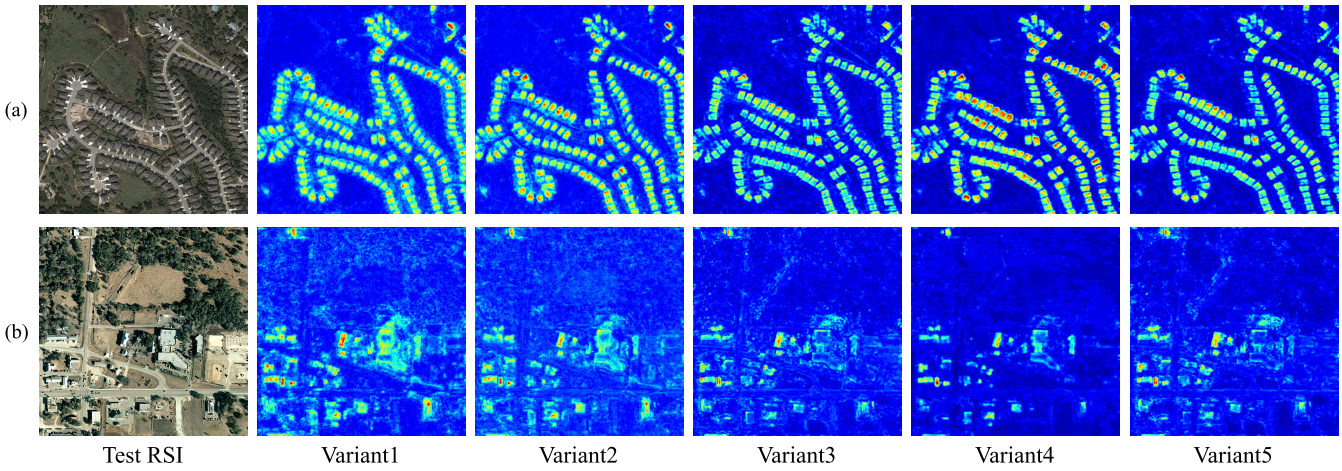



Fig. 6. Feature map of different Meta-CD variants with two test RSIs (a) and (b) in the LEVIR-CD dataset.

PBM, allowing the module to dynamically adjust the threshold for each pixel based on the local context. This enables the model to adaptively set thresholds, thereby improving the accuracy of the binary change map. Table I shows that the network equipped with PBM (Variant 4) significantly outperforms the network using a fixed threshold (Variant 3) across all evaluation metrics, with improvements of 0.36%, 3.14%, and 4.59% in OA, mIoU, and $F1$ score, respectively. The capability of PBM ensures that the model can effectively handle varying levels of change intensity across different regions of an image. This is crucial in remote sensing applications, where the nature of changes can be highly heterogeneous. Thus, PBM helps reduce false positives in stable regions and false negatives in areas with subtle changes. In Fig. 6, we present the feature maps of different variants. Specifically, with the introduction of PBM, the feature maps become clearer and more discriminative, demonstrating the effectiveness of PBM in significantly enhancing change detection capabilities.

### *E. Comparative Experiments*

In this section, we compare the RSCD performance of Meta-CD with mainstream state-of-the-art (SOTA) methods.

1) *BIT [12]:* A transformer-based method that enhances the context information of CNN features using a transformer encoder–decoder network with semantic tokens, and subsequently applies feature differencing to generate the change map.
2) *ChangeFormer [13]:* A transformer-based Siamese network architecture that effectively extracts multiscale long-range detailed information by utilizing a hierarchical transformer encoder and a lightweight MLP decoder.
3) *STANet [23]:* A metric-based Siamese FCN architecture that incorporates a spatial–temporal attention mechanism to enhance feature discrimination.
4) *TinyCD [43]:* A lightweight model that adopts a Siamese U-Net architecture, leveraging low-level features for feature mixing and attention mechanisms in both global spatiotemporal and local spatial contexts.
5) *SNUNet [44]:* A Siamese network based on nested U-Net that employs dense skip connections to preserve high-resolution details and the ensemble channel attention module (ECAM) to refine features.
6) *Changer [45]:* A Siamese encoder–decoder architecture, enhancing the interaction between bitemporal features by introducing interaction layers during the feature extraction process.
7) *SAM-CD [28]:* A large model that utilizes VFM and introduces a task-independent semantic learning branch to model the semantic potential in bitemporal RSI.

Table II presents the performance of different methods across three datasets of varying scales. Upon comparing these methods, we find that Meta-CD excels in multiple key metrics, indicating its high accuracy and stability in RSCD tasks. Specifically, Meta-CD achieves SOTA performance in nearly all metrics. Notably, in terms of the mIoU metric, Meta-CD consistently surpasses the current advanced methods Changeer and SAM-CD on the LEVIR-CD and S2Looking datasets of varying scales. Although Meta-CD slightly underperforms compared to SAM-CD on 50% of the WHU-CD dataset, the difference is marginal, and it maintains strong performance overall. In addition, we observe that the OA levels are quite close, which we attribute to the influence of class balance and sample size on accuracy. In these datasets, there is a significant disparity in the number of samples per class, with fewer samples in the changed category compared to the unchanged category. Consequently, models that perform better across more categories achieve higher OA. Overall, models based on VFM outperform those based on CNN and Transformer architectures, which we attribute to the robust feature extraction capabilities of the underlying model. The results demonstrate that our proposed Meta-CD performs best, validating the rationale behind each module we designed.

Fig. 7 illustrates the performance variation of our method when using different proportions of training samples. It is evident that Meta-CD performs exceptionally well with limited samples, achieving results close to those obtained with larger

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TABLE II

PERFORMANCE OF DIFFERENT RSCD METHODS AND THE PROPOSED META-CD ON THE LEVIR-CD, WHU-CD, AND S2LOOKING DATASETS

| Dataset | Method | 5% | | | 15% | | | 30% | | | 50% | | | 100% | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | OA | mIoU | F1 | OA | mIoU | F1 | OA | mIoU | F1 | OA | mIoU | F1 | OA | mIoU | F1 |
| LEVIR-CD | BIT | 94.90 | 75.35 | 69.60 | 98.59 | 86.96 | 85.93 | 98.12 | 88.53 | 87.84 | 98.22 | 89.70 | 89.30 | 99.13 | 91.59 | 95.45 |
| | ChangeFormer | 97.22 | 75.52 | 70.06 | 98.87 | 89.01 | 88.40 | 98.81 | 88.36 | 87.52 | 98.78 | 88.17 | 87.39 | 99.13 | 91.55 | 95.43 |
| | STANet | 96.79 | 75.98 | 71.23 | 97.99 | 81.52 | 78.88 | 94.90 | 84.04 | 82.35 | 98.19 | 84.84 | 83.05 | 98.84 | 87.57 | 91.40 |
| | TinyCD | 97.97 | 81.85 | 79.38 | 98.77 | 88.28 | 87.55 | 97.95 | 79.39 | 75.71 | 98.54 | 85.69 | 84.33 | 99.13 | 91.61 | 95.46 |
| | SNUNet | 97.78 | 80.02 | 76.80 | 97.87 | 82.41 | 80.28 | 98.20 | 84.27 | 82.64 | 98.17 | 84.68 | 83.23 | 99.14 | 91.67 | 95.50 |
| | Changer | 97.13 | 71.92 | 63.74 | 98.90 | 89.26 | 88.69 | 98.79 | 88.23 | 87.47 | 98.74 | 88.11 | 87.35 | **99.15** | 91.70 | 95.51 |
| | SAM-CD | 97.98 | 81.99 | **79.70** | 98.87 | 88.96 | **90.50** | 99.05 | 90.79 | 89.98 | **99.14** | 90.68 | 90.51 | 99.11 | 91.68 | 95.50 |
| | Meta-CD(Ours) | **98.03** | **82.08** | 79.67 | **99.02** | **90.66** | 90.32 | **99.06** | **91.08** | **90.79** | 99.05 | **90.90** | **90.59** | 99.14 | **91.76** | **95.53** |
| WHU-CD | BIT | 97.52 | 73.21 | 66.40 | 97.89 | 82.49 | 79.92 | 98.42 | 83.65 | 81.55 | 98.36 | 87.71 | 86.65 | 98.29 | 83.72 | 90.46 |
| | ChangeFormer | 98.72 | 83.22 | 80.92 | **99.18** | 87.89 | 86.93 | 98.73 | 89.73 | 89.08 | **99.38** | 92.32 | 92.05 | 99.12 | 90.08 | 94.55 |
| | STANet | 97.16 | 75.73 | 70.47 | 98.79 | 86.24 | 84.88 | 99.07 | 89.31 | 88.63 | 98.95 | 88.25 | 87.38 | 98.63 | 86.89 | 92.57 |
| | TinyCD | 97.78 | 75.61 | 69.66 | 98.83 | 85.04 | 86.38 | 98.83 | 86.93 | 85.77 | 99.14 | 89.57 | 88.90 | 98.70 | 86.34 | 92.20 |
| | SNUNet | 97.38 | 75.65 | 70.13 | 97.83 | 78.36 | 74.18 | 98.00 | 81.03 | 78.15 | 98.71 | 84.84 | 83.04 | 99.16 | 90.45 | 94.77 |
| | Changer | **98.08** | 76.88 | 71.57 | 98.91 | 87.49 | 86.43 | 98.33 | 83.51 | 81.48 | 99.29 | 91.11 | 90.68 | 99.53 | 94.57 | 97.15 |
| | SAM-CD | 98.04 | 88.19 | 87.30 | 98.90 | 90.08 | 87.89 | 98.99 | 91.44 | 91.74 | 99.37 | **94.11** | 93.73 | **99.60** | 95.36 | **97.58** |
| | Meta-CD(Ours) | 97.94 | **90.57** | **90.13** | 98.83 | **90.49** | **90.02** | **99.08** | **93.53** | **92.83** | 99.33 | 93.98 | **93.91** | 99.55 | **95.46** | 97.52 |
| S2Looking | BIT | 98.95 | 63.84 | 72.06 | 98.96 | 67.63 | 76.37 | 99.13 | 69.25 | 78.03 | 99.12 | 70.30 | 79.10 | 99.17 | 70.61 | 79.40 |
| | ChangeFormer | 99.09 | 68.62 | 77.39 | 99.17 | 70.72 | 79.51 | 99.16 | 72.31 | 81.04 | 99.22 | 72.45 | 81.16 | 99.23 | 73.65 | 82.27 |
| | STANet | 97.18 | 58.89 | 66.37 | 97.85 | 65.63 | 72.80 | 98.38 | 65.11 | 72.26 | 98.63 | 65.47 | 72.67 | 98.85 | 66.09 | 73.47 |
| | TinyCD | 98.96 | 65.37 | 73.86 | 99.08 | 69.44 | 78.24 | 99.16 | 71.50 | 80.27 | 99.15 | 71.43 | 80.20 | 99.20 | 72.20 | 80.93 |
| | SNUNet | 98.77 | 62.27 | 70.18 | 99.01 | 67.46 | 76.18 | 99.01 | 67.97 | 76.73 | 99.12 | 70.14 | 78.93 | 99.14 | 69.93 | 78.73 |
| | Changer | 99.07 | 69.44 | **78.24** | 99.09 | 70.93 | 79.73 | **99.17** | 72.83 | **81.53** | 99.19 | 72.87 | **81.56** | 99.25 | 74.49 | **83.03** |
| | SAM-CD | 99.04 | 68.09 | 77.08 | **99.16** | 71.38 | 80.37 | 99.15 | 72.04 | 81.05 | 99.20 | 73.35 | 81.28 | 99.22 | 73.47 | 64.61 |
| | Meta-CD(Ours) | **99.11** | **69.48** | 77.92 | 99.15 | **71.54** | **80.42** | 99.16 | **73.76** | 81.40 | **99.29** | **73.82** | 81.43 | **99.34** | **74.73** | 81.61 |

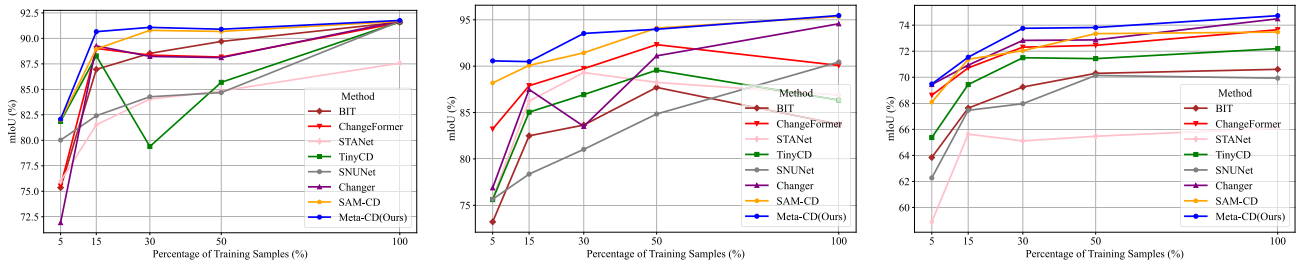



Fig. 7. mIoU at different percentages of training data of (a) LEVIR, (b) WHU, and (c) S2Looking.

sample sizes. This indicates that our model can achieve reliable change detection results without requiring a large number of training samples. As the number of samples increases, we observe a performance decline in some methods, such as TinyCD, which we attribute to potential overfitting. However, Meta-CD consistently maintains a robust level of performance, indicating its scalability and ability to effectively manage overfitting issues. Furthermore, we observed that some methods, such as changer, exhibit performance fluctuations across different sample sizes. In contrast, Meta-CD consistently maintains high performance even with smaller sample sizes, demonstrating its robustness.

Judging from the experimental results in Fig. 7 and Table II, Meta-CD exhibits different performance on various datasets. Taking these three datasets as examples, the model shows good performance when the proportion of training data is 5%. As the amount of data increases, the performance gradually improves, but the growth rate gradually slows down, indicating that the benefits of adding more data are gradually decreasing at this time. Overall, although different datasets vary due to their own characteristics, Meta-CD generally demonstrates that when the proportion of training data is between 15% and 30%, it can achieve a high and stable performance level on multiple evaluation metrics. In practical applications, considering the balance between data acquisition costs and model performance, this range of data volume can be used as a reference and is considered a relatively sufficient amount of data.

Fig. 8 presents a visual comparison of different methods across three datasets. For illustrative purposes, we selected challenging samples with complex architectural features from the testing set. The figure visually compares the performance of various models, showing that even with a limited sample size, the test results closely approximate the GT. To enhance visual clarity, we used different colors and red boxes to highlight key and detailed results, where blue and red indicate missed detections and false positives, respectively. Specifically, Meta-CD excels in avoiding missed and false detections in complex architectural features. Notably, BiT and STANet exhibit significant issues with missed detections. Compared with SAM-CD, which is also based on the VFM method, Meta-CD shows noticeably fewer blue and red pixels. In addition, a common problem among these methods is the presence of scattered false positives in unchanged areas, leading to coarser change maps. In contrast, our method demonstrates superior visual results in regions with limited samples, excelling in boundary handling and small object detection. Observing the red areas adjacent to buildings in the

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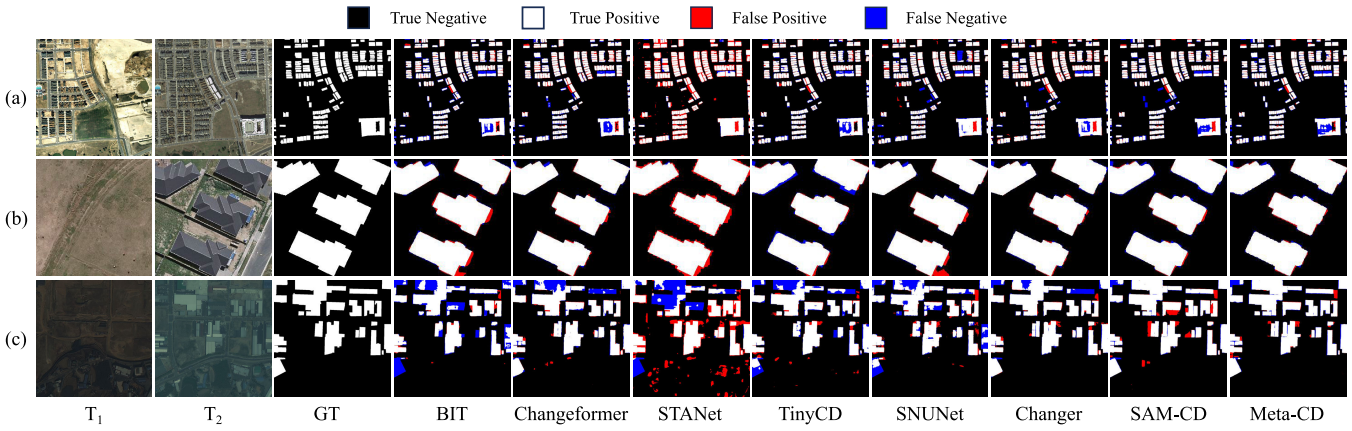



Fig. 8. Visualization results of different methods on (a) LEVIR, (b) WHU, and (c) S2Looking.

TABLE III

COMPUTATIONAL COSTS OF DIFFERENT METHODS ON LEVIR-CD. ONLY THE LEARNABLE PARAMETERS ARE COUNTED

| Methods | Parameters (M) | Flops (G) | Inference (FPS) |
|---|---|---|---|
| BIT | 2.99 | 8.75 | 28.93 |
| ChangeFormer | 13.94 | 5.83 | 18.71 |
| STANet | 13.36 | 48.03 | 1.53 |
| TinyCD | 0.29 | 1.45 | 26.28 |
| SNUNet | 3.01 | 11.73 | 5.96 |
| Changer | 11.39 | 5.96 | 49.31 |
| SAM-CD | 2.49 | 8.60 | 4.59 |
| Meta-CD (Ours) | 13.67 | 10.42 | 5.64 |

TABLE IV

ZERO-SHOT CHANGE DETECTION PERFORMANCE OF DIFFERENT METHODS

| Methods | LEVIR→WHU | | | LEVIR→S2Looking | | |
|---|---|---|---|---|---|---|
| | OA(%) | mIOU(%) | F1(%) | OA(%) | mIOU(%) | F1(%) |
| BIT | 94.35 | 56.97 | 64.96 | 98.76 | 50.26 | 51.41 |
| ChangeFormer | 95.97 | 60.53 | 69.05 | 98.79 | 50.25 | 51.38 |
| STANet | 94.78 | 50.23 | 54.06 | 98.76 | 53.13 | 57.15 |
| TinyCD | 96.18 | 57.85 | 65.06 | 98.83 | 51.24 | 53.34 |
| SNUNet | 95.11 | 62.11 | 71.33 | 98.75 | 50.21 | 51.33 |
| Changer | 95.57 | 47.82 | 48.94 | 98.75 | 49.90 | 50.72 |
| SAM-CD | 97.24 | 77.23 | 74.09 | 98.82 | 52.58 | 55.21 |
| Meta-CD (Ours) | **98.22** | **81.58** | **78.89** | **98.80** | **54.12** | **58.84** |

figure, it is evident that Meta-CD is more sensitive to subtle changes.

Table III presents the computational resource costs of various methods. It is observed that the number of parameters and floating-point operations (FLOPs) significantly impact model performance, but they are not the sole determining factors. Among all the methods compared, TinyCD has the fewest parameters (0.29 M) and FLOPs (1.45 G), while Changer boasts the fastest inference speed (49.31 frames/s); however, their actual performance is not outstanding. In Meta-CD, the FastSAM encoder accounts for the majority of the parameters, with the network module requiring updates occupying 13.67 M parameters. Despite having a relatively high parameter count, its computational efficiency surpasses that of most methods, and its inference speed is moderate. We believe that Meta-CD strikes a good balance between performance and efficiency, and more importantly, it performs well even with minimal data.

### *F. Zero-Shot Performance*

In the RSCD task, different datasets exhibit distinct characteristics and styles, necessitating traditional change detection methods to undergo specialized training for each specific dataset. The LEVIR dataset primarily focuses on building changes in urban environments, the WHU dataset encompasses both urban and rural settings, and the S2Looking dataset includes a wider variety of scenes and change types. Consequently, traditional change detectors often perform poorly when applied across different datasets, as they are typically tailored to datasets with styles similar to their training data. Leveraging the excellent zero-shot transfer capability of the base model, we trained Meta-CD on the LEVIR dataset and tested it on the WHU and S2Looking datasets for validation. The zero-shot detection performance is presented in Table IV.

We observed that traditional RSCD models made numerous erroneous predictions when tested on the WHU dataset. For the mIOU metric, the best performance among traditional RSCD models was achieved by SNUNet at 62.11, whereas our proposed Meta-CD easily reached 81.58, representing a significant improvement over traditional change detectors. Additionally, when testing the trained LEVIR model on the S2Looking dataset, Meta-CD achieved the best zero-shot detection performance. Compared with other models, the improvement by Meta-CD was modest, which we attribute to the highly complex nature of S2Looking. Overall, these results demonstrate the outstanding zero-shot transfer capability of Meta-CD.

## V. CONCLUSION

This study proposes an innovative Meta-CD model, which effectively addresses the issue of limited training samples in RSCD tasks by integrating a pretrained FastSAM model with an additional convolutional adapter and introducing an adaptive PBM. Experimental results demonstrate that the Meta-CD model outperforms current SOTA methods across multiple evaluation metrics, proving that high-accuracy change detection can be achieved with limited annotated data through appropriate model design and fine-tuning strategies. Future work can explore integrating other types of foundational vision models and auxiliary fine-tuning CNNs into the change detection workflow to further optimize model structure and performance. The successful implementation of the Meta-CD

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model offers new perspectives and tools for land cover monitoring, urban planning, and environmental assessment.

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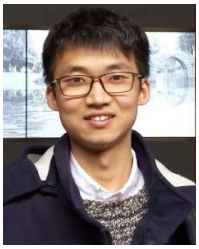
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